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**UNIVERSITY OF INFORMATION
TECHNOLOGY AND SCIENCES**

PROJECT REPORT

For

UniQuery — *“Ask. Explore. Choose.”*

A Smart University Information and Admission Assistant

Prepared By:

Israt Jahan — 0432320005101130

Md. Mehedi Hasan — 0432320005101082

Course: Software Project Design & Development (CSE 416)

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Prepared For:

Dr. Mahfida Amjad Dipa

Senior Teacher

Department of CSE

EXECUTIVE SUMMARY

UniQuery is an intelligent, AI-powered conversational assistant designed to bridge the information gap for high school graduates and admission-seeking students. Navigating university admissions across multiple public and private institutions is often overwhelming due to fragmented information, complex eligibility rules, varying fee structures, and strict deadlines. UniQuery simplifies this critical phase by providing a centralized, interactive "Ask, Explore, Choose" platform.

By leveraging Natural Language Processing (NLP) and dynamic query processing, UniQuery allows students to ask free-form questions regarding eligibility, tuition fees, department rankings, and application deadlines. The system analyzes student profiles (such as GPA and academic background) to deliver personalized recommendations and clear visual comparisons. Developed as part of the Software Project Design and Development course, UniQuery aims to transform university discovery into a seamless, accessible, and stress-free digital experience.

PROJECT BACKGROUND

In higher education, selecting the right university and academic discipline is one of the most pivotal decisions in a student's career. However, secondary and higher secondary students routinely face severe obstacles during the admission season:

Information Fragmentation: Universities disseminate admission circulars across disparate websites, PDF documents, local news outlets, and social media pages.

Static and Cluttered Portals: Traditional higher education websites are often cluttered, difficult to search, and lack personalized guidance for students with specific GPA limitations or budget constraints.

High Guidance Costs & Misinformation: Students frequently rely on informal advice, unofficial social media groups, or costly admission coaching centers, leading to unreliable information and missed deadlines.

With recent advancements in conversational AI and web software engineering, there is an urgent opportunity to deploy an automated, responsive, and user-centric platform that serves as a 24/7 personal academic counselor.

PROBLEM STATEMENT

Core Challenge: "Admission-seeking students lack a unified, intelligent, and real-time guidance system to easily search, evaluate, and compare university options based on their individual academic profiles, financial preferences, and career goals."

Specifically, the problem encompasses three major dimensions:

Complexity in Eligibility Assessment: Calculating aggregate eligibility across different unit

criteria, background subjects, and secondary exam boards is confusing and error-prone for applicants.

Time-Consuming Comparison: Manually comparing tuition fees, campus amenities, faculty credentials, and financial aid opportunities across multiple universities requires dozens of hours of manual research.

Lack of Real-Time Assistance: Admission help desks are often overwhelmed, leaving critical inquiries regarding application deadlines and documentation requirements unanswered during peak admission cycles.

PROPOSED SOLUTION & CORE FEATURES

UniQuery directly addresses these pain points by deploying an integrated web application powered by an intelligent conversational bot. The platform follows a three-stage user workflow: **Ask** (natural language interaction), **Explore** (data visualization and comparison), and **Choose** (data-driven decision making).

1. Conversational AI Assistant (Ask)

Natural Language Interface capable of answering queries regarding admission requirements, fee structures, faculty details, and application deadlines in real time.

2. Smart Eligibility Matcher (Explore)

Algorithm that takes student GPA, background discipline, and budget inputs to generate a tailored list of eligible programs and universities.

3. Side-by-Side Comparison (Choose)

Interactive dashboard comparing multiple universities across parameters like tuition fees, scholarship availability, lab facilities, and campus locations.

4. Centralized Notice & Deadline Tracker

Automated database updating critical admission dates, exam schedules, seat counts, and application portal links to prevent missed deadlines.

TEAM FORMATION & WORK DISTRIBUTION

The UniQuery project is designed and executed by a 2-member engineering team for the course Software Project Design and Development. Responsibilities are strategically distributed to align individual focus with core development domains.

Team Member	Role	Key Responsibilities
Israt Jahan ID: 0432320005101130	System Architect & Developer	• Overall Project Management & System Architecture • AI Model Integration & NLP Logic Implementation • Frontend UI/UX Interface Design & Interactive Modules • Backend API Development & System Integration • Testing, Optimization & Deployment Pipelines
Md. Mehedi Hasan ID: 0432320005101082	Database Engineer & Project Documentarian	• Relational Database Design & Schema Modeling • University Data Gathering, Structuring & Normalization • Database Maintenance, Query Optimization & Security • Comprehensive Software Project Documentation • System Requirement Specifications (SRS) & User Guides